

Node Satellization and Power Law: Historical, Regional, and Digital Evidence for Mexico

Elán Zainos Corona

Fractal Core Research · Tlaxcala, Mexico

elan.zainos.corona@gmail.com

March 2026

DOI: <https://doi.org/10.5281/zenodo.19131327>

SSRN: <https://ssrn.com/abstract=6418778>

GitHub: <https://github.com/Inzainos/The-shadow-Node-Theory>

Pre-print — not peer reviewed. Replication package available.

"If you remain still long enough, you can see the algorithm that moves the universe."

— Zainos Corona, E. (2026). Original synthesis.

Abstract

This paper presents Shadow Node Theory v2.2 (SNT), a formal model of node satellization verified across four historical and regional domains. The central hypothesis posits that when two power nodes orbit in critical proximity, the dominant node satellizes the peripheral one through an algorithm whose dynamics follow a power law $R(t) = a \cdot t^b$ invariant to temporal scale and substrate. The model is verified on four historical cases (Bruges-Antwerp, Toledo-Madrid, Portugal-NW Europe, Tlaxcala-Puebla), an N-body matrix of 32 Mexican federal entities ($b = -0.473$, $R^2 = 0.838$, $p < 0.001$), and a corpus of 230 country pairs from the Maddison Project 2023.

The central finding is that the binary model underestimated Tlaxcala's satellization by a factor of $9.3\times$ by ignoring direct extraction toward Mexico City, which concentrates 89.2% of the composite gradient. Abrupt triggers produce significantly faster satellization than gradual ones — institutional friction is the most important predictor of velocity: domains without friction (biological $b_{\text{mean}} = +1.4$, planetary $b_{\text{mean}} = +1.0$) produce $b > 1$; domains with high friction (sovereign countries $b_{\text{mean}} = -0.098$) produce $b \approx 0$ (Mann-Whitney $p = 1.91 \times 10^{-5}$, $n = 486$). The model includes six falsifiability criteria (RC1–RC6) plus RC8 — the mutual interdependence brake — and a four-step diagnostic protocol for regional policy application.

Keywords: satellization, power law, regional inequality, Mexico, complex systems, leapfrog, Tlaxcala, preferential attachment, development economics, N-body matrix

JEL Codes: O18, O11, R11, R12, C22

1. Introduction

Persistent regional inequality in Mexico — and in particular the case of Tlaxcala as one of the poorest federal entities despite its privileged geographic location — represents a problem that conventional development economics models do not satisfactorily explain. Poverty trap models (Sachs 2005) attribute the persistence of poverty to insufficient capital accumulation. Institutional models (Acemoglu & Robinson 2012) attribute it to extractive institutions. Both approaches capture part of the phenomenon but do not formalize the extraction dynamics between specific nodes within a regional system.

The Shadow Node Theory (SNT) proposes a distinct mechanism: satellization — the process by which a node with greater accumulated advantage extracts the residual productive energy of a peripheral node in critical proximity — whose dynamics follow a power law invariant to temporal scale. The most important empirical finding of the model is that the conventional binary model (which analyzes only the Tlaxcala-Puebla pair) underestimated Tlaxcala's satellization gradient by $9.3\times$ by ignoring that 89.2% of the extraction flows directly toward Mexico City, not toward Puebla. Any regional development strategy targeting only the Tlaxcala-Puebla relationship addresses 10.8% of the actual problem.

2. Theoretical Framework

2.1 The Satellization Hypothesis

When two nodes orbit in critical proximity within the same system, the satellization ratio $R(t) = \text{production_hub}(t) / \text{production_shadow}(t)$ follows a power law $R(t) = a \cdot t^b$ where $b > 0$ indicates active satellization and $b < 0$ indicates convergence or leapfrog. The exponent b measures the velocity of the process: $b > 0.45$ corresponds to accelerated satellization associated with abrupt triggers; $0.1 < b < 0.45$ to gradual satellization; $b < -0.1$ to convergence or leapfrog.

The mechanism is preferential attachment (Barabási & Albert 1999): once a dominant node obtains an initial advantage of 10–15%, the opportunity cost of investing in the historical node becomes mathematically prohibitive. The system self-perpetuates through three suppression vectors: legal (regulations acting as regulatory firewalls), logistic (infrastructure bypass that increases the latency of peripheral node operations), and gravitational (human capital flight toward the larger center of mass).

2.2 Five-Level Node Taxonomy

The SNT v2.2 classifies nodes in five functional levels. Level 0 (Central Hub): maximum preferential attachment inertia; primary function is unidirectional absorption. Level 1 (Secondary Attractors): nodes with sufficient gravitational mass to generate their own preferential attachment fields over immediate peripheries, while simultaneously being drained by the Level 0 hub. Level 2 (Logistic Bypass): structures optimized for flow interception when the Level 0 hub exceeds its maximum load capacity K_{max} . Level 3 (Deep Shadow Nodes): base stratum operating under severe historical gradient satellization; primary candidates for leapfrog. Level E (Exogenous Anomalies): nodes sustained by direct injections from networks external to the national system.

Empirical verification with INEGI 2022 data for the 32 federal entities: Level 0 (Mexico City, 14.8% of national GDP), Level 1 (9 secondary attractors, 41.0%), Level 2 (8 logistic bypass nodes, 20.2%), Level 3

(11 shadow nodes, 16.8%), Level E (3 exogenous anomalies, 4.3%). The distribution follows a power law confirming preferential attachment: $b=-0.473$, $R^2=0.838$, $p<0.001$.

2.3 Trigger Types

Satellization triggers are classified into three categories. Abrupt: datable events with immediate effect (political decree, infrastructure collapse, conquest — e.g., Toledo-Madrid 1561, Bruges-Antwerp c.1490). Gradual: cumulative processes without a central identifiable event (differential industrialization, accumulated colonial advantages — e.g., Tlaxcala-Puebla). Hybrid: datable event whose effect depends on the subsequent diffusion of a technology or infrastructure (e.g., Portugal-Iberian Union 1580 plus Atlantic route diffusion). The corpus of 502 cases confirms that abrupt triggers produce significantly faster satellization than gradual ones (Mann-Whitney $U=24,802$, $p=1.91\times 10^{-5}$).

2.4 Falsifiability Criteria

The model includes eight explicit refutation criteria. The most relevant for the regional domain are: RC6 (Irreversibility) — falsified if a Shadow Node reverses satellization from inside the system without exogenous trigger, with the hub operating normally; RC8 (Mutual Interdependence Brake) — falsified if sovereign-state systems produce sustained $b>0.5$ without exogenous perturbation. The cases of Querétaro ($b=-0.155$, $p<0.01$) and Nuevo León ($b=-0.058$, $p<0.001$) document successful leapfrog via orthogonal dimension without refuting RC6, because both required federal investment and international trade agreements as exogenous triggers.

3. Methodology

3.1 General Design

The study follows a mixed design combining historical case analysis, N-body matrix modeling for the Mexican regional system, and cross-verification with the Maddison 2023 corpus. The unified analytical pipeline: (1) construction of the satellization ratio time series $R(t)$ for each node pair; (2) log-log transformation: $\log(R) = \log(a) + b \cdot \log(t)$; (3) OLS estimation of parameters a and b ; (4) reporting of exponent b , R^2 , and Pearson correlation p -value in logarithmic space; (5) SNT taxonomy classification. For the N-body matrix, we additionally compute the composite gradient — the total satellization force on each node accounting for all higher-level extracting hubs, not just the immediate one.

3.2 Data Sources

Historical cases: Bairoch, Batou & Chèvre (1988) for European urban population 800–1850; Maddison Project Database 2023 (Bolt & van Zanden 2024) for GDP per capita. Mexican system: INEGI National Accounts System of Mexico (SCNM) 2022–2023, state GDP series 1993–2022; Maddison Project proxy estimates for 1940–1993. Country corpus: Maddison Project Database 2023, 169 countries, GDP per capita 1820–2022. All data and scripts available in the replication package at <https://doi.org/10.5281/zenodo.19131327>.

4. Results — Historical Cases

4.1 The Four Foundational Cases

The four original historical cases establish the two-speed taxonomy. Bruges→Antwerp ($b=+0.787$, $R^2=0.987$, $p<0.001$): abrupt trigger — silting of the Zwin Canal (c.1490). In 1300 Bruges had approximately 46,000 inhabitants — four times Antwerp's population; by 1560 the inversion was complete. Toledo→Madrid ($b=+0.687$, $R^2=0.894$, $p<0.001$): abrupt trigger — Philip II's decree (1561). Toledo had 56,270 inhabitants in 1561 versus fewer than 15,000 in Madrid; by 1661 Madrid had 120,000 while Toledo had fallen below 25,000, a population loss of 55%. Portugal→NW Europe ($b=+0.290$, $R^2=0.550$, $p<0.01$): hybrid trigger — Iberian Union (1580) plus Atlantic route diffusion. Tlaxcala→Puebla ($b=+0.184$, $R^2=0.567$, $p<0.001$): gradual trigger — differential colonial accumulation.

The ratio between abrupt and gradual classes is approximately $5.9\times$. This bimodal distribution is the central empirical finding of SNT v1.0, confirmed and refined across the 502-case corpus.

4.2 The Composite Gradient of Tlaxcala: Correcting the Binary Model

The most important finding of SNT v2.2 is the systematic underestimation of the binary model. The binary model measured $w_{ij}(\text{Tlaxcala} \rightarrow \text{Puebla}) = 26.2\text{k MXN}$ (2022 GDP per capita differential). The N-body matrix reveals $w_{ij}(\text{Tlaxcala} \rightarrow \text{Mexico City, long-range}) = 216.8\text{k MXN}$. Total composite gradient: 243.0k MXN. The binary model underestimated Tlaxcala's satellization by a factor of $9.3\times$. The 89.2% of extraction flows directly to Mexico City bypassing the Puebla intermediary. The composite gradient equation for a Level 3 node:

$$\text{Total_flow} = w(\text{Level3} \rightarrow \text{Level1}) + w(\text{Level3} \rightarrow \text{Level0})$$

This is the critical refinement of the original binary model. The binary model error is not that it is incorrect — it captures one level of the extraction chain and ignores the others. The N-body matrix is the step from the simple differential equation to the coupled system of equations.

5. Results — Mexican Regional System

5.1 N-Body Power Law

Verification with INEGI 2022 data for the 32 federal entities confirms that the system follows preferential attachment: $b=-0.473$, $R^2=0.838$, $p<0.001$. The power-law fit on GDP per capita ranked by entity: $f(\text{rank}) = 396.8 \times \text{rank}^{(-0.473)}$. The Pareto distribution is confirmed: the 30.6% of entities at Level 0 + Level 1 (10 of 32) concentrate 55.8% of national GDP; the 34.4% at Level 3 generate only 16.8% of national GDP despite being the most numerous group.

5.2 Trajectory Vectorization (1940–2022)

Analysis of eight entities reveals two natural groups. Satellization ($b>0$): Chiapas ($b=+0.229$, $R^2=0.839$), Oaxaca ($b=+0.176$), Guerrero ($b=+0.176$), Veracruz ($b=+0.181$), Tlaxcala ($b=+0.147$), Puebla ($b=+0.116$). Convergence ($b<0$): Querétaro ($b=-0.155$, $R^2=0.782$, $p<0.01$) and Nuevo León ($b=-0.058$, $R^2=0.935$, $p<0.001$). These two represent the first documented cases of successful leapfrog within the national system — Querétaro via aerospace manufacturing, Nuevo León via independent export manufacturing — both via orthogonal dimensions where Mexico City had no accumulated advantage.

5.3 Leapfrog Mechanism

Mathematical simulations confirm that it is impossible to close the gap by competing linearly. The accumulated advantage of upper-level hubs is insurmountable on the plane of physical infrastructure. The only viable rupture is asymptotic: jumping directly to a dimension where the accumulated advantage does not yet exist. Verified cases of successful leapfrog at national scale: Estonia (1991, digital infrastructure), Rwanda (fiber optic tech hub), Ireland (fiscal and technological strategy surpassing UK per capita GDP), Medellín (from most violent to innovation hub in 15 years).

6. Results — Maddison 2023 Country Corpus

Verification with 230 country pairs from the Maddison Project 2023 produces the most important finding on model limits: $b_{\text{mean}} = -0.098$ for the complete set. Sovereign countries do not definitively satellize each other despite GDP per capita gaps of up to $100\times$. Political sovereignty acts as a braking mechanism that prevents definitive satellization — mechanically identical to what operates in predator-prey ecological relationships ($b_{\text{mean}} = +0.102$). Both cases produce $b \approx 0$ because complete node extinction would destroy the hub.

Abrupt triggers produce $b_{\text{mean}} = +0.552$ versus gradual $b_{\text{mean}} = -0.013$ (Mann-Whitney $U = 24,802$, $p = 1.91 \times 10^{-5}$, $n = 486$ cases across all domains). This result has remained stable across three successive corpus expansions from 57 to 114 to 502 cases — the standard of empirical robustness for an inductive finding.

7. Diagnostic Protocol for Regional Policy

The SNT v2.2 is prescriptive as well as descriptive. Four steps for application in any real system with structured data.

Step 1 — Level Classification: collect node production data. Verify whether the distribution follows a power law (log-log fit, $R^2 > 0.7$, $p < 0.05$). Classify in the five-level taxonomy based on relative position in the distribution.

Step 2 — Composite Gradient Calculation: calculate w_{ij} for each hub extracting from the node. Calculate the composite gradient if there are multiple hubs. The most common error is measuring only the gradient toward the immediate hub. The Tlaxcala case shows the long-range gradient can be $8.3\times$ larger than the direct one.

Step 3 — Event Horizon Estimation: fit the historical trajectory. If $b > 0$ (satellization), estimate t_{horizon} — the point where residual energy falls below the leapfrog activation energy under the current extraction rate. If $b < 0$ (convergence), identify the mechanism and verify its sustainability.

Step 4 — Orthogonal Dimension Identification: search for dimensions where the hub has not invested in 5–10 years, where the node has a measurable initial advantage, and where the dimension has preferential-attachment potential. Verify it does not require hub-controlled infrastructure. Design the intervention in three phases: (1) address the critical deficiency at minimum cost; (2) build capacity in the new dimension without triggering the immune response; (3) execute the leapfrog when the window is open.

8. Limitations

The empirical corpus has several limitations. Data uncertainty of $\pm 20\%$ for pre-1820 historical estimates. Small N in foundational historical cases (4 original cases). Low R^2 in the Portugal case (0.123 — oscillation from exogenous perturbations, particularly the Brazilian gold cycle 1700–1750). Possible reverse causality in the digital case. State GDP series 1940–1993 are proxy estimates constructed using the Maddison Project, not direct INEGI data. The Triple-Resolution Model (Micro/Meso/Macro) has partial operationalization — the Micro and Macro modules require independent empirical validation. The complete replication package is available at <https://doi.org/10.5281/zenodo.19131327> for independent verification.

9. Conclusions

Shadow Node Theory v2.2 demonstrates that regional satellization follows a verifiable power law with INEGI data. The binary model underestimated Tlaxcala's satellization by 9.3 $\times$. The first cases of successful leapfrog within the national system are documented: Querétaro and Nuevo León via orthogonal dimension. The corpus of 230 country pairs confirms that political sovereignty is a satellization braking mechanism as effective as ecological interdependence. Abrupt triggers produce satellization 5.9 $\times$ faster than gradual ones.

The most important implication is not theoretical but practical. The 89.2% of Tlaxcala's problem does not lie in its relationship with Puebla but in its relationship with Mexico City. Any regional development strategy that ignores that asymmetry solves only a marginal fraction of the problem. The leapfrog toward AI-native economy — operating in the cloud, not on the ground — is the first dimension in recent history where the accumulated advantage of dominant nodes does not directly apply to new entrants.

"The satellization algorithm is predictable. What is predictable can be intervened."

References

- Acemoglu, D. & Robinson, J.A. (2012). Why Nations Fail. Crown Publishers.
- Barabási, A.L. & Albert, R. (1999). Emergence of scaling in random networks. Science, 286(5439), 509-512.
- Bolt, J. & van Zanden, J.L. (2024). Maddison Project Database 2023. University of Groningen. <https://www.rug.nl/ggdc/historicaldevelopment/maddison/>
- Brezis, E.S. & Krugman, P.R. (1993). Leapfrogging in international competition: A theory of cycles in national technological leadership. American Economic Review, 83(5), 1211-1219.
- Holland, J.H. (1995). Hidden Order: How Adaptation Builds Complexity. Addison-Wesley.
- INEGI (2022). PIB per cápita por entidad federativa. Sistema de Cuentas Nacionales de México. <https://www.inegi.org.mx/temas/pib/>
- Sachs, J. (2005). The End of Poverty. Penguin Press.

Watts, D.J. & Strogatz, S.H. (1998). Collective dynamics of small-world networks. *Nature*, 393(6684), 440-442.

Zainos Corona, E. (2026). Shadow Node Theory v2.2 — Replication Package. Zenodo. <https://doi.org/10.5281/zenodo.19131327>